ATRIA INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
BENGALURU

FIELD VISIT DATA COLLECTION FORMAT for MULTIDISCIPLINARY PROJECT

PROJECT TITLE: "SAHAYAK - BRIDGING THE TECHNOLOGICAL GAP IN EDUCATION AND NUTRITION"

SDG: SECTOR 7 - Education and Edtech Initiatives

GUIDE NAME: Prof. Y MARY ANGEL

TEAM LEAD's DETAILS BELOW:					
STUDENT NAME	**USN**	**DEPT**	**SECTION**	**PHONE#**	**ATRIA EMAIL ID**
HARSHINI D	1AT24CS072	CSE	B	9591205258	1at24cs072@atria.edu
TEAM MEMBER's DETAILS BELOW:					
STUDENT NAME	**USN**	**DEPT**	**SECTION**	**PHONE#**	**ATRIA EMAIL ID**
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FIELD VISIT DETAILS BY HARSHINI D :

1. **Field Visit Details**

Date(s) of Visit	Sector Visited (Identify from the SDG List)	Whom did you meet? (Name, Role of the person, Phone #, Email ID (if applicable)	Address / Location where you met the person(s)	Is the person working in an organization? (Y/N)
10.09.25	Education &EdTech Initiatives	Teachers and students Teacher: Pranesh K Deshpande - 8095521147	Government High School, Kailashpuram	Yes
10.09.25	Education &EdTech Initiatives	Teachers and students Teacher: Nayaz - 9902319924	Good Sheperd Covent High School	Yes

2. **Person(s) Visited Interaction Summary (Data Collection)**

Person(s) Name	Role of the Person	Sector Visited (Organized / Un-Organized)	Mode of Interaction (Personal Meetings must have photo / video evidence)
Mr. Pranesh K Deshpande	English Teacher	Organized	Personal Meeting
Mr. Nayaz	Computer Applications Teacher	Organized	Personal Meeting

Key Issues Highlighted:

1. By Teachers :
Government High School –

- At government school, technology is used in a very limited way. The only major use of technology is a Karnataka government website called SATS (sts.karnataka.gov.in) is used to track student attendance. This system helps to make sure students who are present receive their midday meals and also keeps their information secure and also this is used for issuing the TC to the students.
- The teachers even spoke about the lack of support from the parents towards their children education and future .
- Students show disinterests in learning new things .
- However, the school is not using technology for other purposes. Teachers work is not being made easier , all the assignments and notes are made individually by themselves, and technology is not used

to improve learning for students. The school mainly sticks to traditional ways of teaching and managing things.

Good Sheperd Convent High School –

- Although this is a private school yet they don't have any major tech implementations since they follow state syllabus curriculum .
- Unlike government school even they use the Karnataka government website called SATS (sts.karnataka.gov.in) to track student attendance and store the details related to students and for the teachers they use a mobile app with face recognition that tracks teachers attendance.

2. By Students :

Government High School -

- **Underutilized Technology:** The school has modern tools like computers and smart boards, but they are not used frequently for teaching.
- **Limited Computer Education:** Computer classes are held **only twice a month**, and the curriculum is very basic, focusing only on how to operate a computer. This means students are not being taught about advanced topics or practical applications.
- **Knowledge Gap vs. Digital Engagement:** Students also addressed that their peers are heavily involved in social media and online games and has lack of knowledge about technology and AI.

Good Sheperd Convent High School –

- Since the syllabus remains same as that of government school only difference is that frequent computer classes are being scheduled but only the basics of computer are being thought.
- Students do use platforms like youtube , chatgpt etc for their studies because of lack of understanding in classrooms taught by teachers.
- Few students even addressed the lack of opportunities and facilities about their studies.

3. **Observations & Data Collected**

Observed Challenges/Issues	Approximate Quantitative Data	Photos/Sketches /Documents Attached
1. Underutilization of technology for teaching and learning.	**Frequency of Computer Classes: 2 classes per month.** **Approximate Time Lost:** A full school year has ~220 school days. With only 2 computer classes a month, students are missing out on at least 80% of potential technology education time.	PFA images below

2. Significant gap between formal education and students digital engagement.	**Frequency of Tech Topics in Curriculum:** 0 times per month (beyond basic computer operation). **Estimated Digital Engagement:** A high percentage of students (likely 100%) own a smartphone or have access to one and are on social media/gaming platforms.	PFA images below
3. Lack of parental support and student disinterest.	Quantitative data on this is not provided. This is more qualitative.	PFA images below
4 . Lack of opportunities and resources beyond the basic curriculum.	**Frequency of Extracurricular Tech Activities/Workshops:** 0 times per month.	PFA images below

4. Problem Statement

Problem Statement
The real problem is despite having some technology, both the government and private schools struggle to provide students with a relevant and modern education. They primarily use a government website for attendance and records, but they fail to integrate technology into the daily learning process. As a result, students are not being taught essential digital skills, creating a significant gap between their classroom education and their active digital lives on social media and with online games. This issue is compounded by a lack of parental support and student disinterest, which further hinders academic progress and future opportunities. This creates a serious gap. Students are growing up in a high-tech world, but their schools are leaving them behind with a low-tech education. In short **,**The schools have the tools but aren't teaching students how to build a better future with them.

5. Alignment with UN Sustainable Development Goals (SDGs)

Relevant SDG(s)	Justification (How the problem aligns)
Sector 7 : Education and Edtech Initiatives – • Rural and Government Schools	This problem directly goes against three key UN goals: • **Goal 4: Quality Education.** The schools aren't using their technology to give students a proper, modern education. They have computers and smartboards, but they're not using them to teach the skills students need for the future. • **Goal 8: Decent Work.** By not teaching important tech skills, the schools are failing to prepare students for the jobs of tomorrow. This makes it harder for them to find good work and build a career. • **Goal 9: Innovation.** The schools have modern tools but aren't using them in an innovative way. They're

	stuck in old teaching methods, which prevents them from building a more advanced and skilled generation

❖ **Attachments – Media files :**

- **Questions asked during field visit –**

SDG Sector 7 : EDUCATION & EdTech INITIATIVES
Government schools -

- For Teachers -

1) Beyond teaching the curriculum, how do you see tech helping you in your own professional life, for ex: in preparing lessons or tracking student progress

2) What kind of tech (Ex - comp's, projectors etc) & how often is it used during reg classes?

3) What are the challenges faced when trying to use technology for teaching? [Slow internet, broken devices, or lack of training]

4) Do you feel that students learn better when they are working on a tablet / comp alone, or when they are working in a small group? why?

5) If you could have a simple dashboard, what would be the top three things you'd want it to show you about your student's learning progress?

6) What kind of digital content would be most useful? Do you prefer videos, interactive quizzes, or something else?

7) What improvements do you wish to see in future

- For Students -

1. Do you have a smartphone or computer at home? for studies?

2. What is your fav thing to do on a comp or a phone?

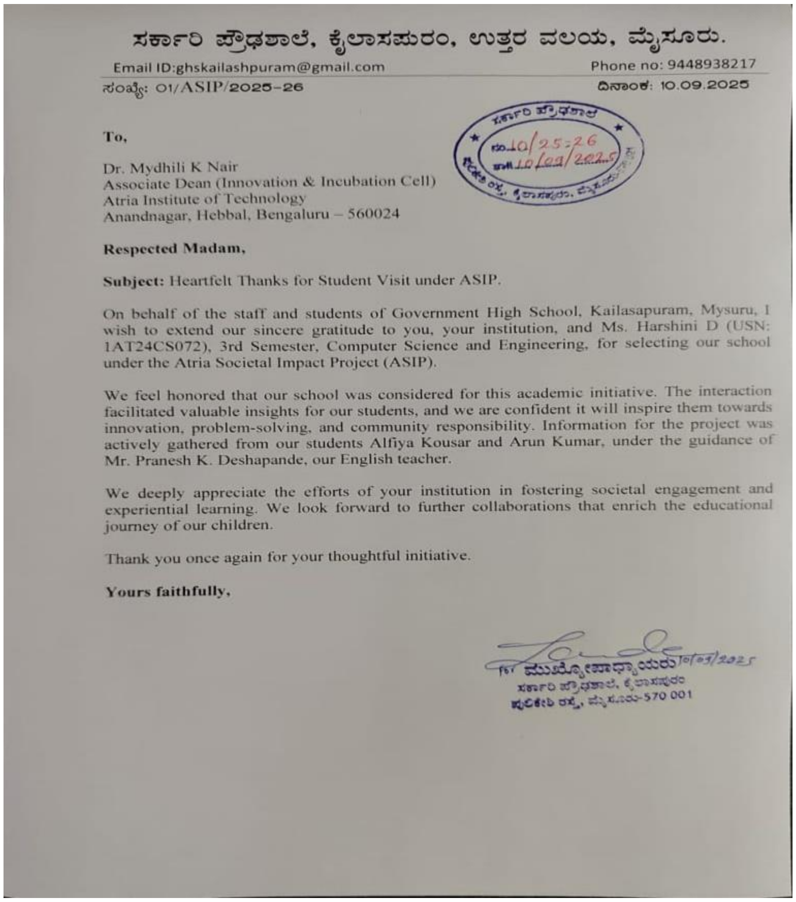
3. Do you like comp's? why or why not?

4. What kind of games or apps do you like to play? What if your school's app was like that?

5. If you wish to learn something beyond curriculum, what would be the first thing you would want to learn & why?

6) What's the hardest part about learning after school? What would make it easier? [not having books, no one to help with hw etc]

❖ **Supporting Documents :**



ಸರ್ಕಾರಿ ಪ್ರೌಢಶಾಲೆ, ಕೈಲಾಸಪುರಂ, ಉತ್ತರ ವಲಯ, ಮೈಸೂರು.

Email ID:ghskailashpuram@gmail.com Phone no: 9448938217

ಸಂಖ್ಯೆ: 01/ASIP/2025-26 ದಿನಾಂಕ: 10.09.2025

To,

Dr. Mydhili K Nair
Associate Dean (Innovation & Incubation Cell)
Atria Institute of Technology
Anandnagar, Hebbal, Bengaluru – 560024

Respected Madam,

Subject: Heartfelt Thanks for Student Visit under ASIP.

On behalf of the staff and students of Government High School, Kailasapuram, Mysuru, I wish to extend our sincere gratitude to you, your institution, and Ms. Harshini D (USN: 1AT24CS072), 3rd Semester, Computer Science and Engineering, for selecting our school under the Atria Societal Impact Project (ASIP).

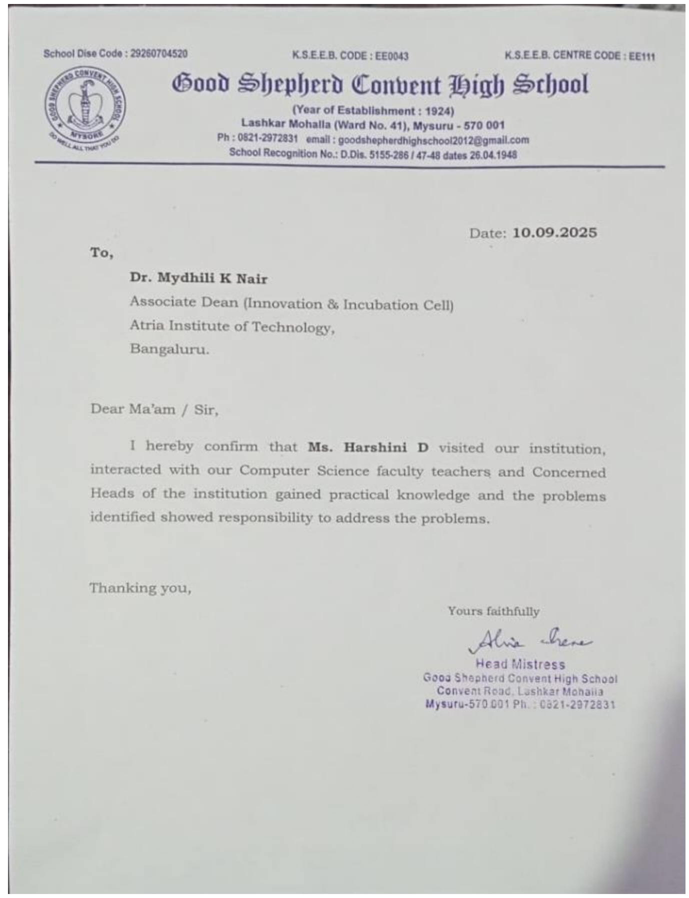
We feel honored that our school was considered for this academic initiative. The interaction facilitated valuable insights for our students, and we are confident it will inspire them towards innovation, problem-solving, and community responsibility. Information for the project was actively gathered from our students Alfiya Kousar and Arun Kumar, under the guidance of Mr. Pranesh K. Deshapande, our English teacher.

We deeply appreciate the efforts of your institution in fostering societal engagement and experiential learning. We look forward to further collaborations that enrich the educational journey of our children.

Thank you once again for your thoughtful initiative.

Yours faithfully,

ಮುಖ್ಯೋಪಾಧ್ಯಾಯರು 10/09/2025
ಸರ್ಕಾರಿ ಪ್ರೌಢಶಾಲೆ, ಕೈಲಾಸಪುರಂ
ಪುಲಿಕೇಶಿ ರಸ್ತೆ, ಮೈಸೂರು-570 001



School Dise Code : 29260704520 K.S.E.E.B. CODE : EE0043 K.S.E.E.B. CENTRE CODE : EE111

Good Shepherd Convent High School

(Year of Establishment : 1924)
Lashkar Mohalla (Ward No. 41), Mysuru - 570 001
Ph : 0821-2972831 email : goodshepherdhighschool2012@gmail.com
School Recognition No.: D.Dis. 5155-286 / 47-48 dates 26.04.1948

Date: **10.09.2025**

To,

Dr. Mydhili K Nair
Associate Dean (Innovation & Incubation Cell)
Atria Institute of Technology,
Bangaluru.

Dear Ma'am / Sir,

I hereby confirm that **Ms. Harshini D** visited our institution, interacted with our Computer Science faculty teachers and Concerned Heads of the institution gained practical knowledge and the problems identified showed responsibility to address the problems.

Thanking you,

Yours faithfully

Head Mistress
Good Shepherd Convent High School
Convent Road, Lashkar Mohalla
Mysuru-570 001 Ph. : 0821-2972831

Overall Insights Gained from Field Visit :

❖ **Technology is Underused:** While both schools have some technology, like computers and smartboards, they're not using it effectively for daily teaching. The main tech use is a single government website for attendance and records.

❖ **Students are Digitally Disconnected from Learning:** Students are very active on social media and online games but are not being taught how to use technology for education or career skills. They feel they're not getting enough opportunities to learn what they need for the future.

❖ **The Problem is Deeper than Technology:** Teachers are also facing challenges, including a lack of parental support and students who aren't interested in learning. This makes it harder for them to use new teaching methods, even if the tools were available.

❖ **No Difference Between Schools:** Both the government and private schools follow the same state curriculum, which limits their ability to use technology more widely. This means students in both types of schools are getting a similar, basic tech education.

FIELD VISIT DETAILS BY MOHAMMED ALI FAROOQ -

1. **Field Visit Details**

Date(s) of Visit	Sector Visited (Identify from the SDG List)	Whom did you meet? (Name, Role of the person, Phone #, Email ID (if applicable)	Address / Location where you met the person(s)	Is the person working in an organization? (Y/N)
17/09/2025	• Education and EdTech Initiatives • Food & Nutrition Security	Mrs.Mujeeba.G(Head Mistress) ph: 7829179833	GMPS School, Kadugodi	Y

2. **Person(s) Visited Interaction Summary (Data Collection)**

Person(s) Name	Role of the Person	Sector Visited (Organized / Un-Organized)	Mode of Interaction (Personal Meetings must have photo / video evidence)
MUJEEBA G	Head Mistress	Organized	Personal meeting with photos and videos
MANJUNATH A H	Assistant Teacher	Organized	Personal meeting with photos and videos
Key Issues Highlighted: Lack of enough staff, broken cctv cameras, no security, lack of infrastructure, less use of technology			

3. **Observations & Data Collected**

Observed Challenges/Issues	Approximate Quantitative Data	Photos/Sketches/ Documents Attached
Less technology usage	**Frequency of Computer Classes: 2 classes per month.** **Approximate Time Lost:** A full school year has ~220 school days. With only 2 computer classes a month, students are missing out on at least 80% of potential technology education time.	Photos attached

Mid day meal nutrition not being tracked properly	Quantitative data on this is not provided. This is more qualitative.	Photos attached
Students looking dull and weak due to lack of good health	Quantitative data on this is not provided. This is more qualitative.	Photos attached

4. Problem Statement Draft (Tentative)

Draft Problem Statement
Manual tracking of student attendance, meals, and performance is time-consuming and error-prone. Students lack centralized access to learning resources and timely feedback. Nutritional intake and health monitoring are inefficient, leading to unnoticed deficiencies. Teachers struggle to identify learning gaps or at-risk students quickly. Health alerts and communication with doctors/NGOs are limited, and student achievements are not digitally organized.

5. Alignment with UN Sustainable Development Goals (SDGs)

Relevant SDG(s)	**Justification (How the problem aligns)**
• **SDG 3: GOOD HEALTH AND WELL-BEING(Healthcare and Public Health)** • **SDG 2: ZERO HUNGER (Food And Nutrition Security)** • **SDG 4: QUALITY EDUCATION (Education and EdTech Initiatives)**	1. **Healthcare and Public Health:** Health monitoring, alerts, and doctor/NGO connect help detect and address health issues like anemia, low BMI, or growth problems, promoting student well-being. 2. **Food And Nutrition Security:** By tracking mid-day meals, nutrition, and calories, the system ensures students get adequate, healthy food, contributing to hunger eradication and improved nutrition. 3. **Education and EdTech Initiatives:** Centralized learning resources, performance analytics, and learning gap alerts improve access to education, help struggling students, and enhance learning outcomes.

❖ **Attachments – Media files :**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Kyosan India Private Limited

FIELD VISIT DETAILS By Varsha C-

1. Field Visit Details

Date(s) of Visit	Sector Visited (Identify from the SDG List)	Whom did you meet?	Address / Location where you met the person(s)	Is the person working in an organization? (Y/N)
15/09/2025	• Education and EdTech Initiatives	Vijaylaxmi (9901737737)	BBMP High School,Ganganagar	Y

2. Person(s) Visited Interaction Summary (Data Collection)

Person(s) Name	Role of the Person	Sector Visited (Organized / Un-Organized)	Mode of Interaction (Personal Meetings must have photo / video evidence)
Vijaylaxmi	Teacher	Organized	Personal meeting with photos
Venkatesh	Teacher	Organized	Personal meeting with photos

Key Issues Highlighted:

Lack of enough staff, broken cctv cameras, no security, lack of infrastructure, less use of technology

3. Observations & Data Collected

Observed Challenges/Issues	Approximate Quantitative Data	Photos/Sketches/ Documents Attached
Less technology usage	**Frequency of Computer Classes: 2 classes per month.** **Approximate Time Lost:** A full school year has ~220 school days. With only 2 computer classes a month, students are missing out on at least 80% of potential technology education time.	Photos attached
Low Student Engagement	Quantitative data on this is not provided. This is more qualitative.	Photos attached
Outdated Teaching Methods	Quantitative data on this is not provided. This is more qualitative.	Photos attached

4. Problem Statement

Problem Statement
Government schools in India face challenges in providing good education because they mostly use old teaching methods, have outdated curriculum, and don't use enough technology. This makes it hard for students to learn important digital skills and apply what they learn in class to real life, making them less prepared for today's tech-driven world.

5. Alignment with UN Sustainable Development Goals (SDGs)

Relevant SDG(s)	Justification (How the problem aligns)
• **Goal 4: Quality Education**	The schools' failure to utilize technology effectively hinders students' access to quality education, leaving them unprepared for the future.
• **Goal 8: Decent Work and Economic Growth**	By not imparting essential tech skills, schools compromise students' future employability and career prospects, impacting economic growth.

❖ **Attachments – Media files :**

CURRENT STATE OF EDUCATION:

- Traditional teaching methods dominate, with limited incorporation of technology.
- Curriculum might be outdated, lacking relevance to modern industry needs.
- Students struggle to connect classroom learning to real-life applications.

EDTECH OPPORTUNITIES:

- Introducing digital tools and platforms could enhance engagement, accessibility, and quality of education.
- Teachers need training and support to effectively integrate technology into their teaching practices.
- Potential for edtech initiatives to address infrastructure gaps, such as providing digital devices or internet connectivity.

STUDENT NEEDS:

- Students require digital literacy skills to succeed in a tech-driven world.
- Personalized learning experiences and adaptive technology can help address individual learning needs.
- Interactive and immersive learning experiences can increase student engagement and motivation.

FIELD VISIT DETAILS BY NOOR E SABA, JAVERIA ANJUM, AYESHA BEGUM:

1. **Field Visit Details**

Date(s) of Visit	Sector Visited (Identify from the SDG List)	Whom did you meet? (Name, Role of the person, Phone #, Email ID (if applicable)	Address / Location where you met the person(s)	Is the person working in an organization? (Y/N)
13/12/25	EDUCATION&EDTECHINTIATIVES	TEACHERSAND STUDENTS	GPVT,KANNADTAMIL MODEL PRIMARY SCHOOL	YES

2. **Person(s) Visited Interaction Summary (Data Collection)**

Person(s) Name	Role of the Person	Sector Visited (Organized / Un-Organized)	Mode of Interaction (Personal Meetings must have photo / video evidence)
MRS KRISHNA VENI	HM	ORGANIZED	PERSONALMEETING
MRSPREMA	ENGLISH TEACHER	ORGANIZED	PERSONALMEETING

Key Issues Highlighted: Student information is scattered across registers, government portals, and verbal communication, leading to delays and data mismatch.

- **Health-related information (allergies, illness history) is not available at the time of meal distribution or emergencies.**
- **Parents receive little to no understandable communication due to language barriers and lack of digital access.**
- **Teachers spend significant non-academic time on record maintenance instead of teaching.**
- **Existing government platforms focus on compliance, not student well-being or personalization.**

3. **Observations & Data Collected**

Observed Challenges/Issues	Approximate Quantitative Data	Photos/Sketches/Documents Attached

Manual student record maintenance across attendance, health, and meals	3–4 different registers per class;~20–30 minutes/day per teacher spent on data entry	PFA images below
No individual-level health tracking during midday meals	100%meals distributed without allergy or medical checks	PFA images below
Language barrier in existing digital systems	Parents informed only during monthly meetings ;emergency communication delay: 1–2 hours	PFA images below
No centralized student profile	Student data duplicated across atleast 2 platforms + paper records	PFA images below

4. Problem Statement

Problem Statement
Despite the availability of basic digital infrastructure in government schools, there is no unified, student-centric system that integrates academic records, health data, nutrition information, and parent communication in an accessible and multilingual manner. Critical student information such as medical conditions, food allergies, attendance patterns, and parental contacts remains fragmented across manual records and isolated portals, resulting in inefficiencies, safety risks, and exclusion of parents due to language and digital barriers. This lack of an integrated, intelligent platform prevents timely decision-making, increases teacher workload, and limits the overall effectiveness of government education and welfare schemes, thereby affecting student well-being and learning outcomes.

5. Alignment with UN Sustainable Development Goals (SDGs)

Relevant SDG(s)	Justification (How the problem aligns)
SECTOR 7: EDUCATION AND EDTECH INITIATIVES RURAL AND GOVERNMENT SCHOOLS	This problem directly goes against three key UN goals: ●Goal 4: Quality Education. The schools aren't using their technology to give students a proper, modern education. They have computers and smartboards, but they're not using them to teach the skills students need for the future. ● Goal 8: Decent Work. By not teaching important tech skills, the schools are failing to prepare students for the jobs of tomorrow. This makes it harder for them to find good work and build a career. ● Goal 9: Innovation. The schools have modern tools but aren't using them in an innovative way. They're stuck in old

	teaching methods, which prevents them from building a more advanced and skilled generation

❖ **Attachments – Media files :**

IDEA / SOLUTION :

INITIAL IDEA –

The Idea:

- **Smart Mid-Day Meal Monitoring System**
  - Digital Meal Register (via QR Scan)
  - Menu Nutrition Tracker (calories & nutrients)
  - Feedback Feature (meal quality & hygiene)

- **Develop a new, simple, and hyper-localized mobile app for the school that goes beyond just attendance. This app would act as a digital hub for both teachers and students.**

- **How it Works:**
  - **Teacher-to-Student Modules:** Teachers can upload daily notes, assignments, and links to educational videos (like YouTube channels) directly to the app. This makes their work easier by creating a reusable digital library.
  - **Gamified / Creative Learning:** The app could have built-in quizzes and challenges related to the day's lessons. Students would earn points or badges for completing tasks, addressing their love for online games in a constructive way.
  - **Parent Portal:** Parents could log in to see their child's progress, view notes, and get alerts, directly addressing the lack of parental involvement.

Why It's Unique: Instead of trying to force students to use a complex, one-size-fits-all platform, this app is designed specifically for the school's needs. It's built to be simple and "sticky" by using gamification, making it more appealing than traditional educational software.

For the government schools the idea would remain similar as mentioned above but with few additional features such as:

- Free availability of the text books (can be accessible offline)
- All the content and tutorials provided in app would be available in the local language due to language barrier or exposure in some rural areas.

- **Health & Care Integration**
  - Health Card Integration (height, weight, BMI, Hb level, etc.)
  - Health Alerts (low BMI, anemia, etc.)
  - Doctor/NGO Connect (health camp requests)

THESE IDEAS CAN BE INTEGRATED TO CREATE A NEW INNOVATION WITH THE HELP OF AI.

FUTURE ENHANCEMENTS & NEXT STEPS –

The Idea : To create a hyperlocal and multilingual focused AI-powered teaching assistant designed to reduce the
burden on teachers in under-resourced, multi-grade classrooms across India.

- How it solves the problem:
  - **Saves Time** by automating lesson planning and material creation.
  - It creates **custom materials** for students of different levels in the same class.
  - It helps bridge the **technology and resource gap** between urban and rural education.

Overall, **it enhances the teaching and** improves classroom learning, and makes education more equal and engaging for every child in India.

How does the idea have a positive impact on the society?

- **Improved Child Health and Nutrition:** By monitoring meals and health metrics, the system ensures children receive adequate nutrition, reducing malnutrition and related health issues.
- **Enhanced Educational Outcomes:** Centralized learning resources, performance analytics, and early identification of learning gaps help students perform better academically.
- **Efficient School Management:** Digital tracking of attendance, meals, health, and achievements streamlines school operations, benefiting teachers, parents, and students alike.